

Amazon S3 Access Point

To Begin with the Lab

Summary of the Lab

In this lab, an S3 bucket was created and two folders, App1 and App2, were added inside it. The bucket's access control was then delegated to **S3 Access Points** through a bucket policy, so access was controlled by access point policies instead of the bucket policy itself. Two IAM users, user-1 and user-2, were created and given basic S3 console permissions, including the ability to list access points and open them. Two access points, AP1 and AP2, were created with policies that allowed each user to upload objects only to their assigned folder. In the final test, user-1 could use AP1 and upload to App1, while user-2 could use AP2 and upload to App2.

Understanding Amazon S3 Access Point

An **S3 Access Point** is a separate access layer for an S3 bucket that lets you define different permissions for different users or applications. It exists because bucket policies can become hard to manage when many users need different access patterns. With access points, each user or app can get its own entry point and its own policy, which improves security, simplifies permission management, and scales better. Access points can also be limited by network location, such as a VPC or internet access, depending on how they are configured.

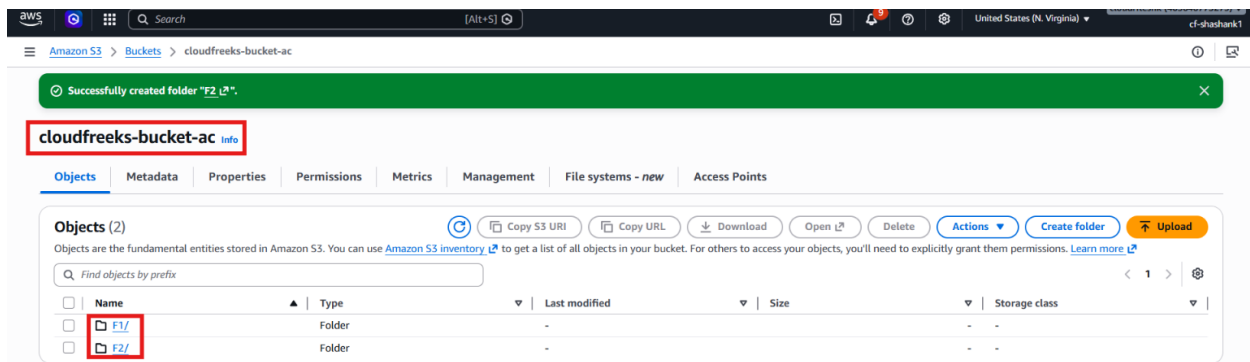
Example:

A bucket has two folders, App1 and App2. user-1 should upload only to App1, and user-2 should upload only to App2. Instead of writing one large bucket policy for both users, you create AP1 and AP2, each with its own policy, so each user can access only the folder meant for them.

Lab Steps

Step 1 – Create the bucket and folders

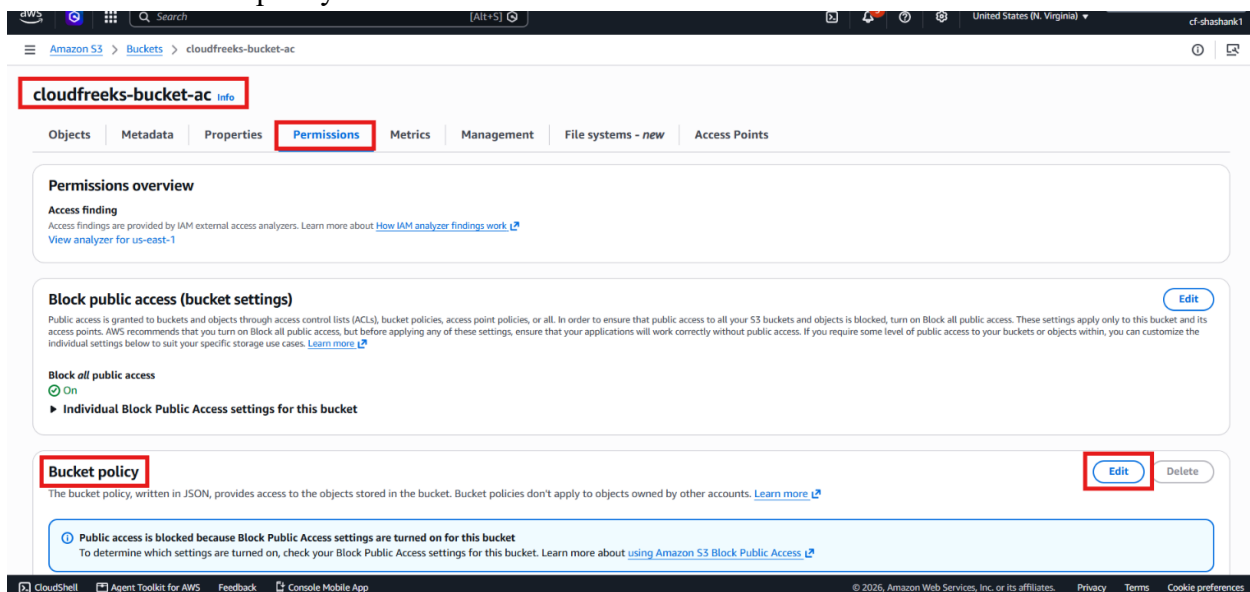
- Open **Amazon S3**.
- Create a new bucket named cloudfreeks-bucket-ac.
- Inside the bucket, create two folders: F1 and F2.



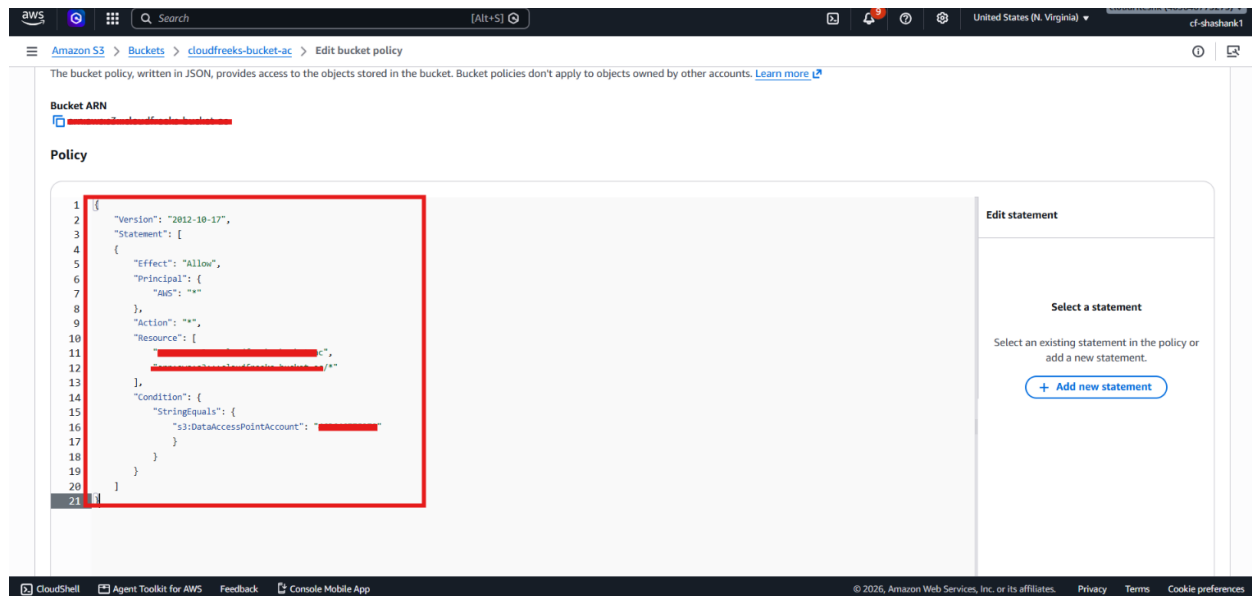
- Confirm that the bucket and folders are ready for the access point setup.

Step 2 – Delegate access control to access points

- Open the bucket **Permissions** tab.
- Edit the bucket policy.



- Paste the policy that delegates control to access points.
- It can be found in the Resources folder.



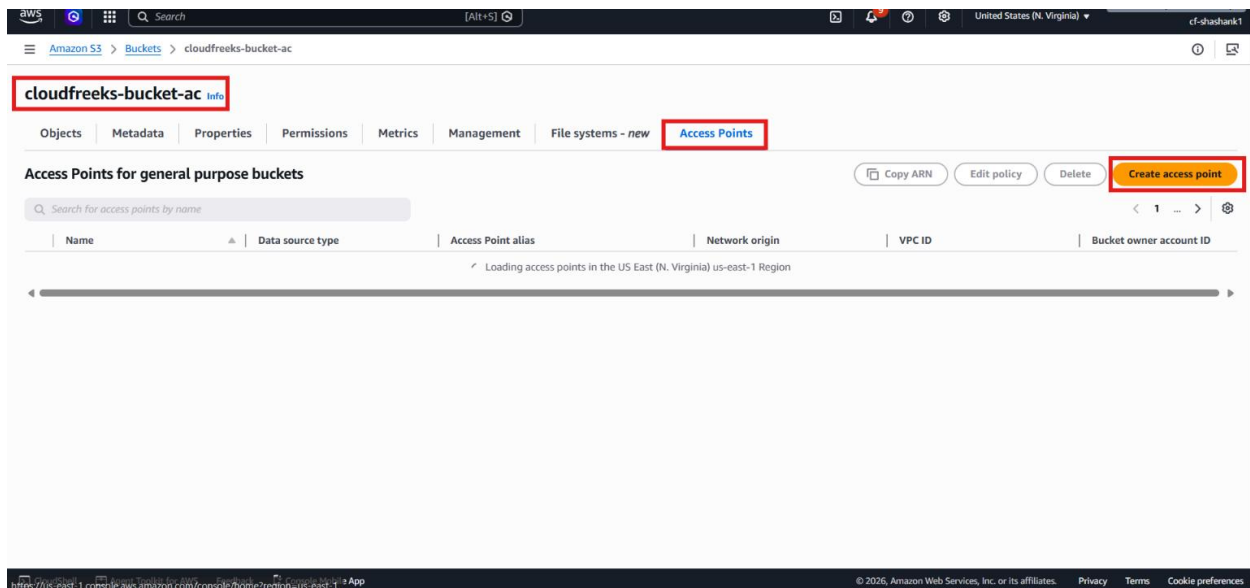
- Replace the bucket ARN and account ID with your own values.
- Save the policy so access will be managed through access points.

Step 3 – Create IAM users

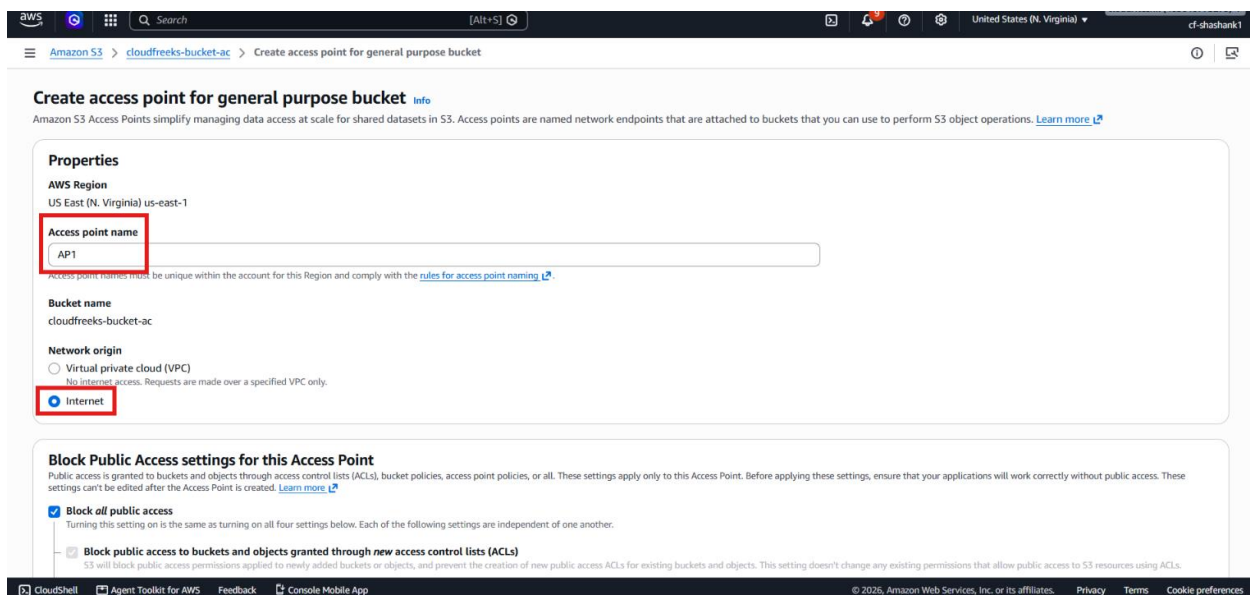
- Open **IAM**.
- Create two users: **amit** and **ravi**.
- Give both users console access.
- Do not attach any permissions yet.
- Sign in with each user and confirm they cannot access S3 objects or access points by default.

Step 4 – Create Access Point for amit

- Return to the S3 bucket.
- Open **Access Points**.
- Click **Create Access Point**.



- Name it: ap1.
- Set it to internet access mode.



- Add a policy that allows amit to use the access point.
- Allow s3:PutObject only into the F1 folder.

Access Point policy - optional

The Access Point policy, written in JSON, provides access to the objects stored in the bucket from this Access Point. Access Point policies don't apply to objects owned by other accounts. [Learn more](#)

Access Point aliases can't be used in the Access Point policy. [Learn more](#)

Access Point ARN

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "allow",
6       "Principal": {
7         "AWS": "*"
8       },
9       "Action": "s3:PutObject",
10      "Resource": "arn:aws:s3:::cloudfreeks-bucket-ac/*"
11    }
12  ]
13 }
```

Edit statement **Remove**

Add actions

Choose a service

Search services

Included

S3

Available

AI Operations

AMP

API Gateway

API Gateway V2

Step 5 – Create Access Point for ravi

- Create another access point named ap2.

Create access point for general purpose bucket [info](#)

Amazon S3 Access Points simplify managing data access at scale for shared datasets in S3. Access points are named network endpoints that are attached to buckets that you can use to perform S3 object operations. [Learn more](#)

Properties

AWS Region

US East (N. Virginia) us-east-1

Access point name

ap2

Access point names must be unique within the account for this Region and comply with the [rules for access point naming](#).

Bucket name

cloudfreeks-bucket-ac

Network origin

☐ Virtual private cloud (VPC)
No internet access. Requests are made over a specified VPC only.

☒ Internet

Block Public Access settings for this Access Point

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. These settings apply only to this Access Point. Before applying these settings, ensure that your applications will work correctly without public access. These settings can't be edited after the Access Point is created. [Learn more](#)

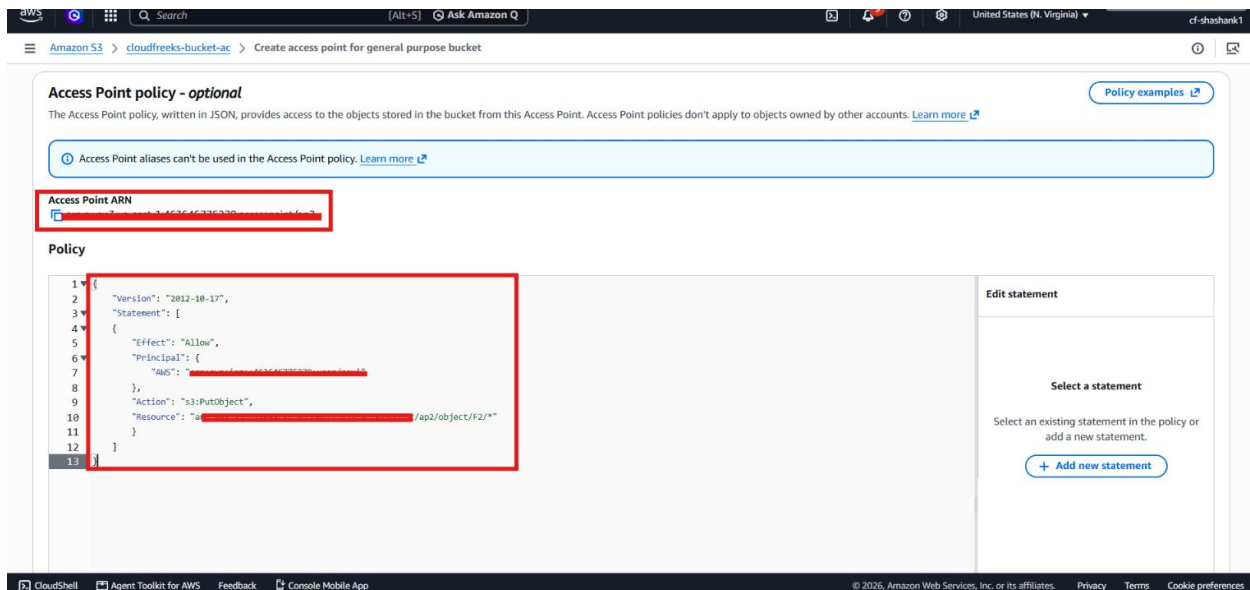
☒ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

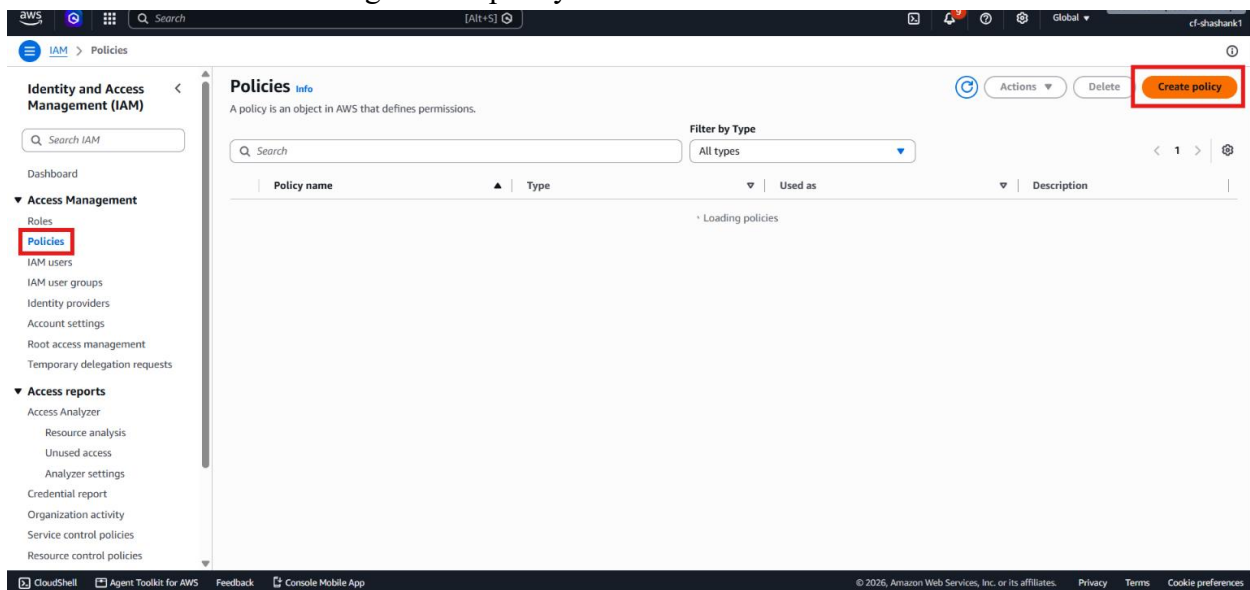
- Attach a policy that allows ravi to use it.
- Allow s3:PutObject only into the F2 folder.



- Save the access point.

Step 6 – Give IAM permissions to use access points

- Create a customer-managed IAM policy for console access.



- Allow:
 - s3:ListAccessPoints
 - s3:GetAccessPoint
 - s3:ListBucket

Step 1
Specify permissions
Step 2
Review and create

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "Statement1",  
6       "Effect": "Allow",  
7       "Action": [  
8         "s3:ListAccessPoints",  
9         "s3:GetAccessPoint",  
10        "s3:ListBucket"  
11      ],  
12      "Resource": "*"   
13    }  
14  ]  
15 }
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

- Attach the policy to both amit and ravi.

Identity and Access Management (IAM)

amit [Info](#) [Delete](#)

Summary

ARN: [arn:aws:iam::463646376330:user/amit](#)

Created: July 21, 2026, 12:50 (UTC+05:30)

Console access: [Enabled without MFA](#)

Last console sign-in: [Today](#)

Access key 1: [Create access key](#)

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

☐ Policy name	Type	Attached via
☐ AmazonS3ReadOnlyAccess	AWS managed	Directly

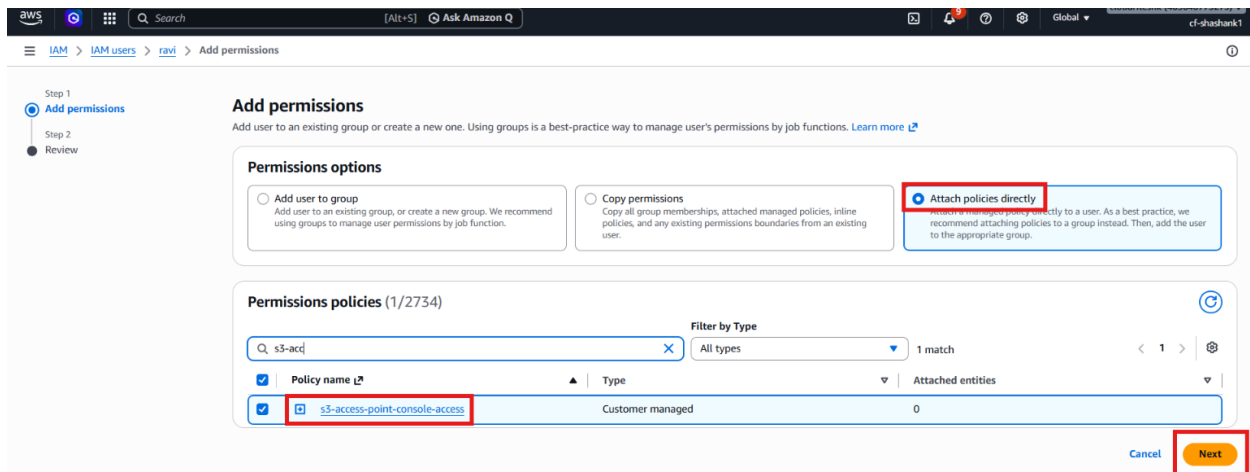
► **Permissions boundary** (not set)

▼ **Generate policy based on CloudTrail events**

[Add permissions](#)

[Add permissions](#)

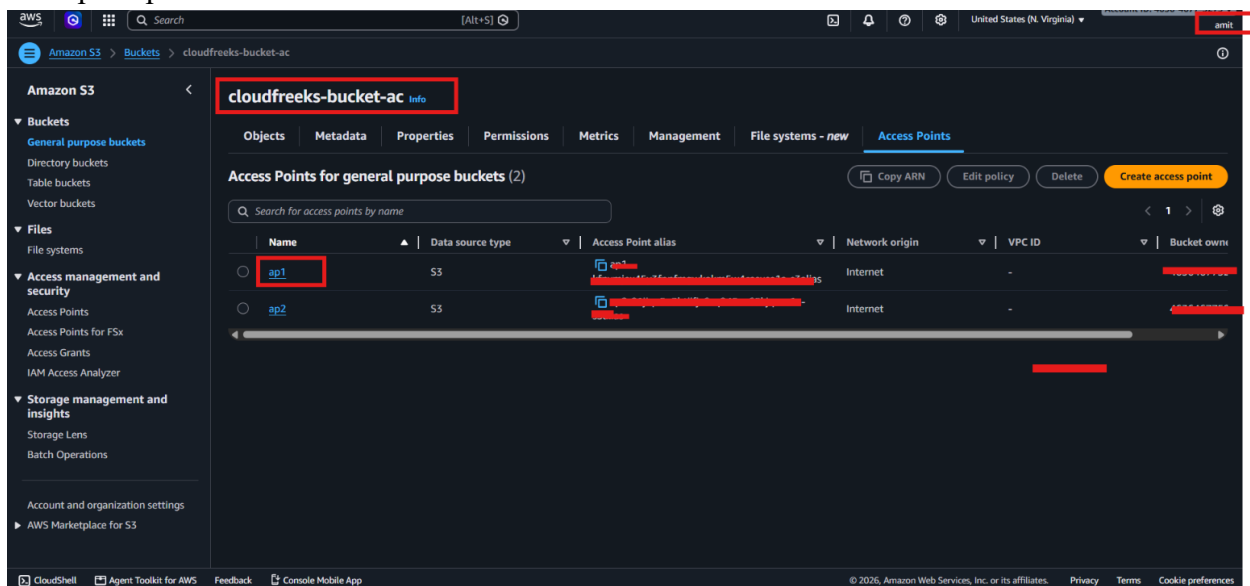
[Create inline policy](#)



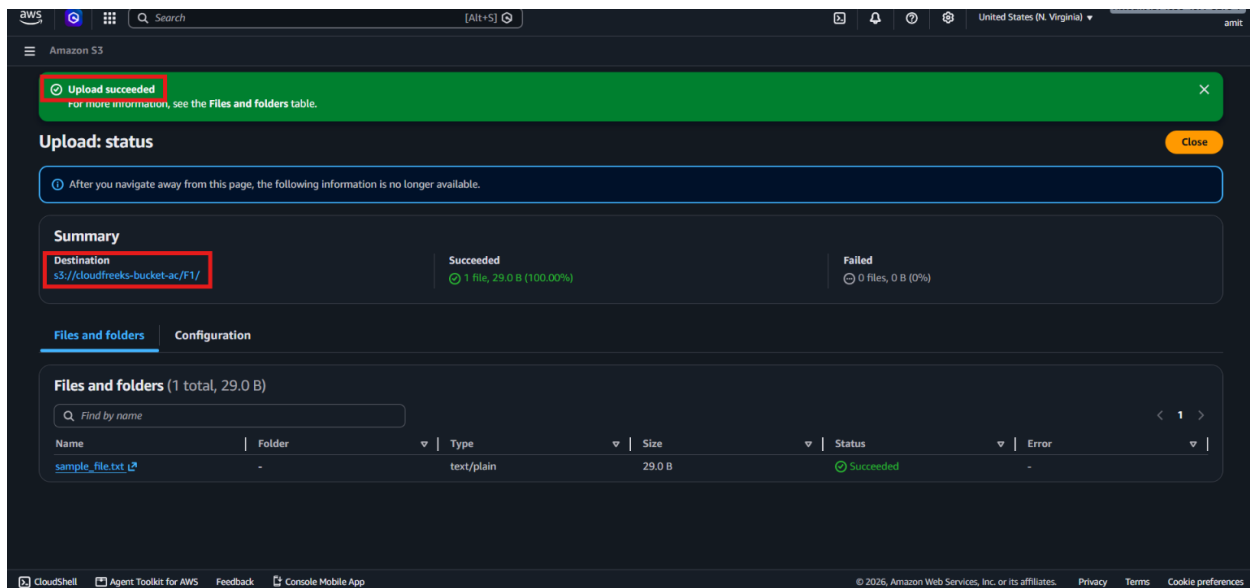
- Refresh the S3 console and confirm that the access points are visible.

Step 7 – Final test

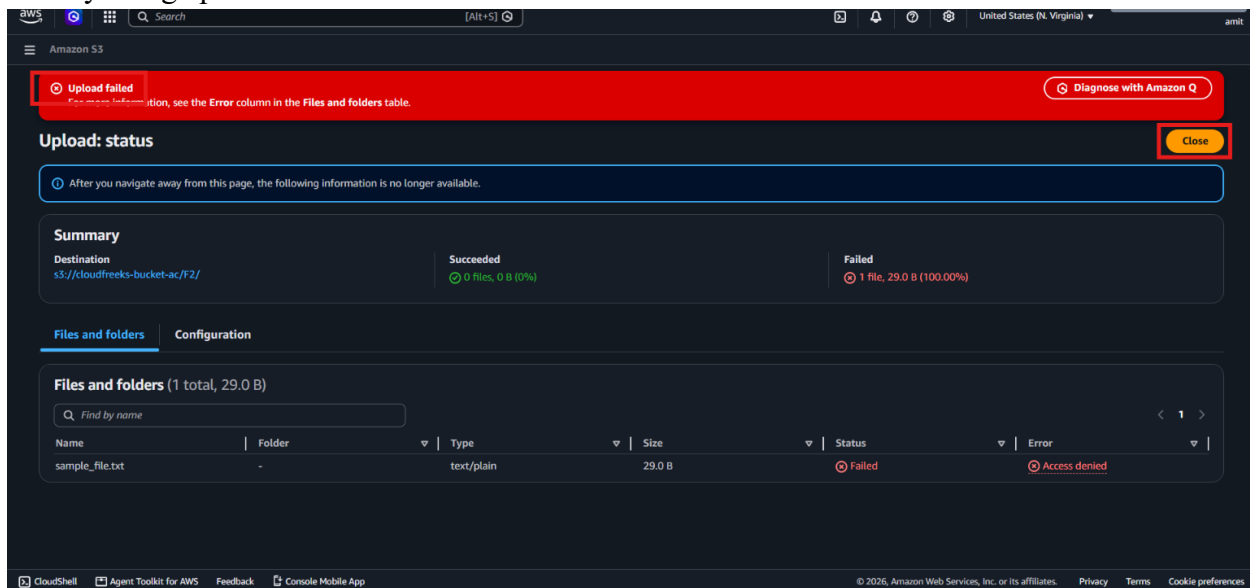
- Log in as amit.
- Open ap1.



- Upload a file and confirm it goes into the F1 folder.



- Try using ap2 and confirm access is denied.



- Log in as ravi.
- Open ap2.
- Upload a file and confirm it goes into the F2 folder.
- Try using ap1 and confirm access is denied.

Verification

- Verified that the bucket and folders were created successfully.
- Verified that bucket access was delegated to access points.
- Verified that user-1 and user-2 had no S3 access before permissions were added.
- Verified that AP1 and AP2 worked with separate policies.
- Verified that user-1 could upload only to App1.
- Verified that user-2 could upload only to App2.

- Verified that IAM permissions were needed to list and open access points in the S3 console.